

Proposition 2.45. *If a finite-dimensional CW complex X is a $K(G, 1)$, then the group $G = \pi_1(X)$ must be torsionfree.*

This applies to quite a few manifolds, for example closed surfaces other than S^2 and $\mathbb{RP}^2$, and also many 3-dimensional manifolds such as complements of knots in S^3 .

Proof: If G had torsion, it would have a finite cyclic subgroup $\mathbb{Z}_m$ for some $m > 1$, and the covering space of X corresponding to this subgroup of $G = \pi_1(X)$ would be a $K(\mathbb{Z}_m, 1)$. Since X is a finite-dimensional CW complex, the same would be true of its covering space $K(\mathbb{Z}_m, 1)$, and hence the homology of the $K(\mathbb{Z}_m, 1)$ would be nonzero in only finitely many dimensions. But this contradicts the fact that $H_n(\mathbb{Z}_m)$ is nonzero for infinitely many values of n . $\square$

Reflecting the richness of group theory, the homology of groups has been studied quite extensively. A good starting place for those wishing to learn more is the textbook [Brown 1982]. At a more advanced level the books [Adem & Milgram 1994] and [Benson 1992] treat the subject from a mostly topological viewpoint.

Mayer-Vietoris Sequences

In addition to the long exact sequence of homology groups for a pair (X, A) , there is another sort of long exact sequence, known as a **Mayer-Vietoris sequence**, which is equally powerful but is sometimes more convenient to use. For a pair of subspaces $A, B \subset X$ such that X is the union of the interiors of A and B , this exact sequence has the form

$$\begin{aligned} \cdots \rightarrow H_n(A \cap B) \xrightarrow{\Phi} H_n(A) \oplus H_n(B) \xrightarrow{\Psi} H_n(X) \xrightarrow{\partial} H_{n-1}(A \cap B) \rightarrow \cdots \\ \cdots \rightarrow H_0(X) \rightarrow 0 \end{aligned}$$

In addition to its usefulness for calculations, the Mayer-Vietoris sequence is also applied frequently in induction arguments, where one might know that a certain statement is true for A , B , and $A \cap B$ by induction and then deduce that it is true for $A \cup B$ by the exact sequence.

The Mayer-Vietoris sequence is easy to derive from the machinery of §2.1. Let $C_n(A + B)$ be the subgroup of $C_n(X)$ consisting of chains that are sums of chains in A and chains in B . The usual boundary map $\partial: C_n(X) \rightarrow C_{n-1}(X)$ takes $C_n(A + B)$ to $C_{n-1}(A + B)$, so the $C_n(A + B)$'s form a chain complex. According to Proposition 2.21, the inclusions $C_n(A + B) \hookrightarrow C_n(X)$ induce isomorphisms on homology groups. The Mayer-Vietoris sequence is then the long exact sequence of homology groups associated to the short exact sequence of chain complexes formed by the short exact sequences

$$0 \rightarrow C_n(A \cap B) \xrightarrow{\varphi} C_n(A) \oplus C_n(B) \xrightarrow{\psi} C_n(A + B) \rightarrow 0$$

where $\varphi(x) = (x, -x)$ and $\psi(x, y) = x + y$. The exactness of this short exact sequence can be checked as follows. First, $\text{Ker } \varphi = 0$ since a chain in $A \cap B$ that is zero as a chain in A (or in B) must be the zero chain. Next, $\text{Im } \varphi \subset \text{Ker } \psi$ since $\psi\varphi = 0$. Also, $\text{Ker } \psi \subset \text{Im } \varphi$ since for a pair $(x, y) \in C_n(A) \oplus C_n(B)$ the condition $x + y = 0$ implies $x = -y$, so x is a chain in both A and B , that is, $x \in C_n(A \cap B)$, and $(x, y) = (x, -x) \in \text{Im } \varphi$. Finally, exactness at $C_n(A + B)$ is immediate from the definition of $C_n(A + B)$.

The boundary map $\partial: H_n(X) \rightarrow H_{n-1}(A \cap B)$ can easily be made explicit. A class $\alpha \in H_n(X)$ is represented by a cycle z , and by barycentric subdivision or some other method we can choose z to be a sum $x + y$ of chains in A and B , respectively. It need not be true that x and y are cycles individually, but $\partial x = -\partial y$ since $\partial(x + y) = 0$, and the element $\partial\alpha \in H_{n-1}(A \cap B)$ is represented by the cycle $\partial x = -\partial y$, as is clear from the definition of the boundary map in the long exact sequence of homology groups associated to a short exact sequence of chain complexes.

There is also a formally identical Mayer-Vietoris sequence for reduced homology groups, obtained by augmenting the previous short exact sequence of chain complexes in the obvious way:

$$\begin{array}{ccccccc} 0 & \longrightarrow & C_0(A \cap B) & \xrightarrow{\varphi} & C_0(A) \oplus C_0(B) & \xrightarrow{\psi} & C_0(A + B) \longrightarrow 0 \\ & & \downarrow \varepsilon & & \downarrow \varepsilon \oplus \varepsilon & & \downarrow \varepsilon \\ 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{\varphi} & \mathbb{Z} \oplus \mathbb{Z} & \xrightarrow{\psi} & \mathbb{Z} \longrightarrow 0 \end{array}$$

Mayer-Vietoris sequences can be viewed as analogs of the van Kampen theorem since if $A \cap B$ is path-connected, the H_1 terms of the reduced Mayer-Vietoris sequence yield an isomorphism $H_1(X) \approx (H_1(A) \oplus H_1(B)) / \text{Im } \Phi$. This is exactly the abelianized statement of the van Kampen theorem, and H_1 is the abelianization of π_1 for path-connected spaces, as we show in §2.A.

There are also Mayer-Vietoris sequences for decompositions $X = A \cup B$ such that A and B are deformation retracts of neighborhoods U and V with $U \cap V$ deformation retracting onto $A \cap B$. Under these assumptions the five-lemma implies that the maps $C_n(A + B) \rightarrow C_n(U + V)$ induce isomorphisms on homology, and hence so do the maps $C_n(A + B) \rightarrow C_n(X)$, which was all that we needed to obtain a Mayer-Vietoris sequence. For example, if X is a CW complex and A and B are subcomplexes, then we can choose for U and V neighborhoods of the form $N_\varepsilon(A)$ and $N_\varepsilon(B)$ constructed in the Appendix, which have the property that $N_\varepsilon(A) \cap N_\varepsilon(B) = N_\varepsilon(A \cap B)$.

Example 2.46. Take $X = S^n$ with A and B the northern and southern hemispheres, so that $A \cap B = S^{n-1}$. Then in the reduced Mayer-Vietoris sequence the terms $\tilde{H}_i(A) \oplus \tilde{H}_i(B)$ are zero, so we obtain isomorphisms $\tilde{H}_i(S^n) \approx \tilde{H}_{i-1}(S^{n-1})$. This gives another way of calculating the homology groups of S^n by induction.

Example 2.47. We can decompose the Klein bottle K as the union of two Möbius bands A and B glued together by a homeomorphism between their boundary circles.

Then A , B , and $A \cap B$ are homotopy equivalent to circles, so the interesting part of the reduced Mayer-Vietoris sequence for the decomposition $K = A \cup B$ is the segment

$$0 \rightarrow H_2(K) \rightarrow H_1(A \cap B) \xrightarrow{\Phi} H_1(A) \oplus H_1(B) \rightarrow H_1(K) \rightarrow 0$$

The map Φ is $\mathbb{Z} \rightarrow \mathbb{Z} \oplus \mathbb{Z}$, $1 \mapsto (2, -2)$, since the boundary circle of a Möbius band wraps twice around the core circle. Since Φ is injective we obtain $H_2(K) = 0$. Furthermore, we have $H_1(K) \approx \mathbb{Z} \oplus \mathbb{Z}_2$ since we can choose $(1, 0)$ and $(1, -1)$ as a basis for $\mathbb{Z} \oplus \mathbb{Z}$. All the higher homology groups of K are zero from the earlier part of the Mayer-Vietoris sequence.

Example 2.48. Let us describe an exact sequence which is somewhat similar to the Mayer-Vietoris sequence and which in some cases generalizes it. If we are given two maps $f, g: X \rightarrow Y$ then we can form a quotient space Z of the disjoint union of $X \times I$ and Y via the identifications $(x, 0) \sim f(x)$ and $(x, 1) \sim g(x)$, thus attaching one end of $X \times I$ to Y by f and the other end by g . For example, if f and g are each the identity map $X \rightarrow X$ then $Z = X \times S^1$. If only one of f and g , say f , is the identity map, then Z is homeomorphic to what is called the mapping torus of g , the quotient space of $X \times I$ under the identifications $(x, 0) \sim (g(x), 1)$. The Klein bottle is an example, with g a reflection $S^1 \rightarrow S^1$.

The exact sequence we want has the form

$$(*) \quad \cdots \rightarrow H_n(X) \xrightarrow{f_* - g_*} H_n(Y) \xrightarrow{i_*} H_n(Z) \rightarrow H_{n-1}(X) \xrightarrow{f_* - g_*} H_{n-1}(Y) \rightarrow \cdots$$

where i is the evident inclusion $Y \hookrightarrow Z$. To derive this exact sequence, consider the map $q: (X \times I, X \times \partial I) \rightarrow (Z, Y)$ that is the restriction to $X \times I$ of the quotient map $X \times I \sqcup Y \rightarrow Z$. The map q induces a map of long exact sequences:

$$\begin{array}{ccccccc} \cdots & \xrightarrow{0} & H_{n+1}(X \times I, X \times \partial I) & \xrightarrow{\partial} & H_n(X \times \partial I) & \xrightarrow{i_*} & H_n(X \times I) \xrightarrow{0} \cdots \\ & & \downarrow q_* & & \downarrow q_* & & \downarrow q_* \\ \cdots & \longrightarrow & H_{n+1}(Z, Y) & \xrightarrow{\partial} & H_n(Y) & \xrightarrow{i_*} & H_n(Z) \longrightarrow \cdots \end{array}$$

In the upper row the middle term is the direct sum of two copies of $H_n(X)$, and the map i_* is surjective since $X \times I$ deformation retracts onto $X \times \{0\}$ and $X \times \{1\}$. Surjectivity of the maps i_* in the upper row implies that the next maps are 0, which in turn implies that the maps ∂ are injective. Thus the map ∂ in the upper row gives an isomorphism of $H_{n+1}(X \times I, X \times \partial I)$ onto the kernel of i_* , which consists of the pairs $(\alpha, -\alpha)$ for $\alpha \in H_n(X)$. This kernel is a copy of $H_n(X)$, and the middle vertical map q_* takes $(\alpha, -\alpha)$ to $f_*(\alpha) - g_*(\alpha)$. The left-hand q_* is an isomorphism since these are good pairs and q induces a homeomorphism of quotient spaces $(X \times I)/(X \times \partial I) \rightarrow Z/Y$. Hence if we replace $H_{n+1}(Z, Y)$ in the lower exact sequence by the isomorphic group $H_n(X) \approx \text{Ker } i_*$ we obtain the long exact sequence we want.

In the case of the mapping torus of a reflection $g: S^1 \rightarrow S^1$, with Z a Klein bottle, the interesting portion of the exact sequence $(*)$ is

$$\begin{array}{ccccccc}
0 \longrightarrow H_2(Z) \longrightarrow H_1(S^1) & \xrightarrow{1-g_*} & H_1(S^1) \longrightarrow H_1(Z) \longrightarrow H_0(S^1) & \xrightarrow{1-g_*} & H_0(S^1) \\
\parallel & & \parallel & & \parallel \\
\mathbb{Z} & \xrightarrow{2} & \mathbb{Z} & \xrightarrow{0} & \mathbb{Z}
\end{array}$$

Thus $H_2(Z) = 0$ and we have a short exact sequence $0 \rightarrow \mathbb{Z}_2 \rightarrow H_1(Z) \rightarrow \mathbb{Z} \rightarrow 0$. This splits since $\mathbb{Z}$ is free, so $H_1(Z) \approx \mathbb{Z}_2 \oplus \mathbb{Z}$. Other examples are given in the Exercises.

If Y is the disjoint union of spaces Y_1 and Y_2 , with $f: X \rightarrow Y_1$ and $g: X \rightarrow Y_2$, then Z consists of the mapping cylinders of these two maps with their domain ends identified. For example, suppose we have a CW complex decomposed as the union of two subcomplexes A and B and we take f and g to be the inclusions $A \cap B \hookrightarrow A$ and $A \cap B \hookrightarrow B$. Then the double mapping cylinder Z is homotopy equivalent to $A \cup B$ since we can view Z as $(A \cap B) \times I$ with A and B attached at the two ends, and then slide the attaching of A down to the B end to produce $A \cup B$ with $(A \cap B) \times I$ attached at one of its ends. By Proposition 0.18 the sliding operation preserves homotopy type, so we obtain a homotopy equivalence $Z \simeq A \cup B$. The exact sequence $(*)$ in this case is the Mayer-Vietoris sequence.

A relative form of the Mayer-Vietoris sequence is sometimes useful. If one has a pair of spaces $(X, Y) = (A \cup B, C \cup D)$ with $C \subset A$ and $D \subset B$, such that X is the union of the interiors of A and B , and Y is the union of the interiors of C and D , then there is a relative Mayer-Vietoris sequence

$$\cdots \rightarrow H_n(A \cap B, C \cap D) \xrightarrow{\Phi} H_n(A, C) \oplus H_n(B, D) \xrightarrow{\Psi} H_n(X, Y) \xrightarrow{\partial} \cdots$$

To derive this, consider the commutative diagram

$$\begin{array}{ccccccc}
& & 0 & & 0 & & 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & C_n(C \cap D) & \xrightarrow{\varphi} & C_n(C) \oplus C_n(D) & \xrightarrow{\psi} & C_n(C + D) \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & C_n(A \cap B) & \xrightarrow{\varphi} & C_n(A) \oplus C_n(B) & \xrightarrow{\psi} & C_n(A + B) \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & C_n(A \cap B, C \cap D) & \xrightarrow{\varphi} & C_n(A, C) \oplus C_n(B, D) & \xrightarrow{\psi} & C_n(A + B, C + D) \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
& & 0 & & 0 & & 0
\end{array}$$

where $C_n(A + B, C + D)$ is the quotient of the subgroup $C_n(A + B) \subset C_n(X)$ by its subgroup $C_n(C + D) \subset C_n(Y)$. Thus the three columns of the diagram are exact. We have seen that the first two rows are exact, and we claim that the third row is exact also, with the maps φ and ψ induced from the φ and ψ in the second row. Since $\psi\varphi = 0$ in the second row, this holds also in the third row, so the third row is at least a chain complex. Viewing the three rows as chain complexes, the diagram then represents a short exact sequence of chain complexes. The associated long exact sequence of homology groups has two out of every three terms zero since the first two rows of the diagram are exact. Hence the remaining homology groups are zero and the third row is exact.

The third column maps to $0 \rightarrow C_n(Y) \rightarrow C_n(X) \rightarrow C_n(X, Y) \rightarrow 0$, inducing maps of homology groups that are isomorphisms for the X and Y terms as we have seen above. So by the five-lemma the maps $C_n(A+B, C+D) \rightarrow C_n(X, Y)$ also induce isomorphisms on homology. The relative Mayer-Vietoris sequence is then the long exact sequence of homology groups associated to the short exact sequence of chain complexes given by the third row of the diagram.

Homology with Coefficients

There is an easy generalization of the homology theory we have considered so far that behaves in a very similar fashion and sometimes offers technical advantages. The generalization consists of using chains of the form $\sum_i n_i \sigma_i$ where each σ_i is a singular n -simplex in X as before, but now the coefficients n_i are taken to lie in a fixed abelian group G rather than $\mathbb{Z}$. Such n -chains form an abelian group $C_n(X; G)$, and there is the expected relative version $C_n(X, A; G) = C_n(X; G)/C_n(A; G)$. The old formula for the boundary maps ∂ can still be used for arbitrary G , namely $\partial(\sum_i n_i \sigma_i) = \sum_{i,j} (-1)^j n_i \sigma_i|[\nu_0, \dots, \hat{\nu}_j, \dots, \nu_n]$. Just as before, a calculation shows that $\partial^2 = 0$, so the groups $C_n(X; G)$ and $C_n(X, A; G)$ form chain complexes. The resulting homology groups $H_n(X; G)$ and $H_n(X, A; G)$ are called **homology groups with coefficients in G** . Reduced groups $\tilde{H}_n(X; G)$ are defined via the augmented chain complex $\dots \rightarrow C_0(X; G) \xrightarrow{\varepsilon} G \rightarrow 0$ with ε again defined by summing coefficients.

The case $G = \mathbb{Z}_2$ is particularly simple since one is just considering sums of singular simplices with coefficients 0 or 1, so by discarding terms with coefficient 0 one can think of chains as just finite ‘unions’ of singular simplices. The boundary formulas also simplify since one no longer has to worry about signs. Since signs are an algebraic representation of orientation considerations, one can also ignore orientations. This means that homology with $\mathbb{Z}_2$ coefficients is often the most natural tool in the absence of orientability.

All the theory we developed in §2.1 for $\mathbb{Z}$ coefficients carries over directly to general coefficient groups G with no change in the proofs. The same is true for Mayer-Vietoris sequences. Differences between $H_n(X; G)$ and $H_n(X)$ begin to appear only when one starts making calculations. When X is a point, the method used to compute $H_n(X)$ shows that $H_n(X; G)$ is G for $n = 0$ and 0 for $n > 0$. From this it follows just as for $G = \mathbb{Z}$ that $\tilde{H}_n(S^k; G)$ is G for $n = k$ and 0 otherwise.

Cellular homology also generalizes to homology with coefficients, with the cellular chain group $H_n(X^n, X^{n-1})$ replaced by $H_n(X^n, X^{n-1}; G)$, which is a direct sum of G ’s, one for each n -cell. The proof that the cellular homology groups $H_n^{CW}(X)$ agree with singular homology $H_n(X)$ extends immediately to give $H_n^{CW}(X; G) \approx H_n(X; G)$. The cellular boundary maps are given by the same formula as for $\mathbb{Z}$ coefficients, $d_n(\sum_\alpha n_\alpha e_\alpha^n) = \sum_{\alpha, \beta} d_{\alpha\beta} n_\alpha e_\beta^{n-1}$. The old proof applies, but the following result is needed to know that the coefficients $d_{\alpha\beta}$ are the same as before: